covid | all 53 locations | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

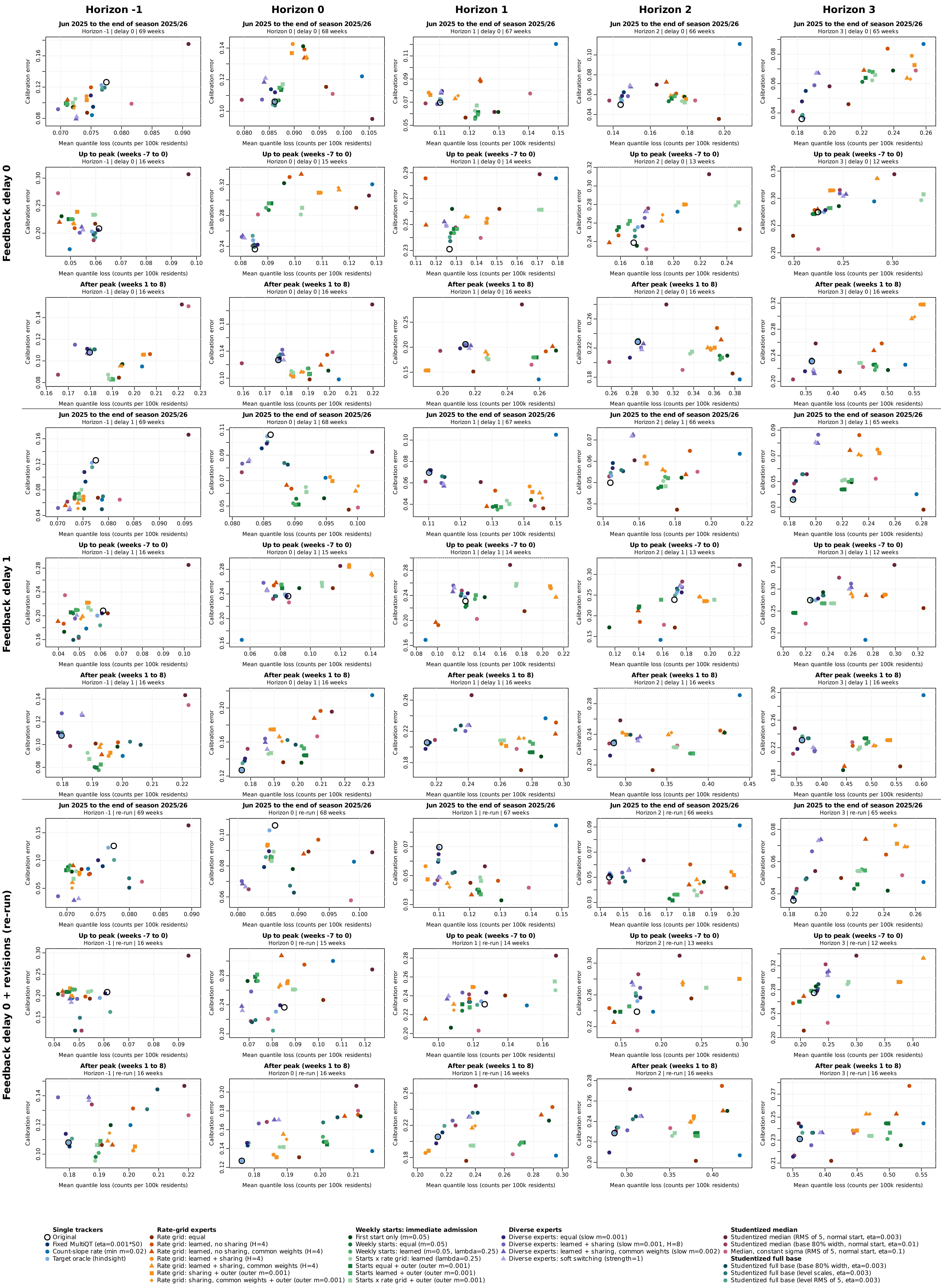
Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

Horizon -1
Horizon 0
Horizon 1
Horizon 2
Horizon 3
Feedback delay 0
Feedback delay 1
Feedback delay 0 + revisions (re-run)
Jun 2025 to the end of season 2025/26
Up to peak (weeks -7 to 0)
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 53 locations
Horizon 0 | delay 0 | 53 locations
Horizon 1 | delay 0 | 53 locations
Horizon 2 | delay 0 | 53 locations
Horizon 3 | delay 0 | 53 locations
Horizon -1 | delay 1 | 53 locations
Horizon 0 | delay 1 | 53 locations
Horizon 1 | delay 1 | 53 locations
Horizon 2 | delay 1 | 53 locations
Horizon 3 | delay 1 | 53 locations
Horizon -1 | re-run | 53 locations
Horizon 0 | re-run | 53 locations
Horizon 1 | re-run | 53 locations
Horizon 2 | re-run | 53 locations
Horizon 3 | re-run | 53 locations
Calibration error
Mean quantile loss (counts per 100k residents)
Single trackers
○ Original
● Fixed MultiQT (eta=0.001*S0)
● Count-slope rate (min m=0.02)
● Target oracle (hindsight)
Rate-grid experts
● Rate grid: equal
● Rate grid: learned, no sharing (H=4)
▲ Rate grid: learned, no sharing, common weights (H=4)
● Rate grid: learned + sharing (H=4)
▲ Rate grid: learned + sharing, common weights (H=4)
■ Rate grid: sharing + outer (outer m=0.001)
◆ Rate grid: sharing, common weights + outer (outer m=0.001)
Weekly starts: immediate admission
● First start only (m=0.05)
● Weekly starts: equal (m=0.05)
● Weekly starts: learned (m=0.05, lambda=0.25)
● Starts x rate grid: learned (lambda=0.25)
■ Starts equal + outer (outer m=0.001)
■ Starts learned + outer (outer m=0.001)
■ Starts x rate grid + outer (outer m=0.001)
Diverse experts
● Diverse experts: equal (slow m=0.001)
● Diverse experts: learned + sharing (slow m=0.001, H=8)
▲ Diverse experts: learned + sharing, common weights (slow m=0.002)
▲ Diverse experts: soft switching (strength=1)
Studentized median
● Studentized median (RMS of 5, normal start, eta=0.003)
● Studentized median (base 80% width, normal start, eta=0.01)
● Median, constant sigma (RMS of 5, normal start, eta=0.1)
Studentized full base
● Studentized full base (base 80% width, eta=0.003)
● Studentized full base (level scales, eta=0.003)
● Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

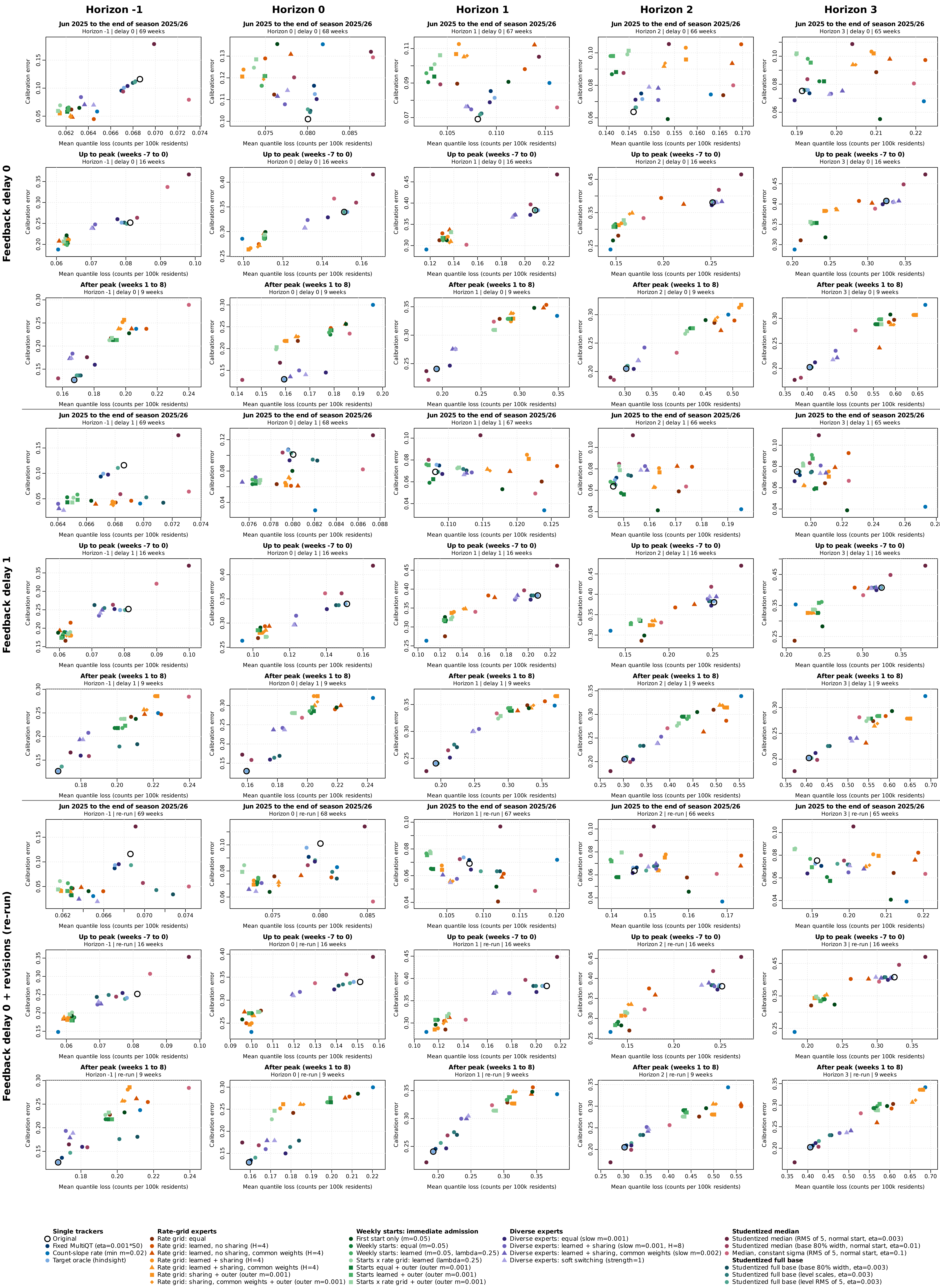
Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | California (06) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Texas (48) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

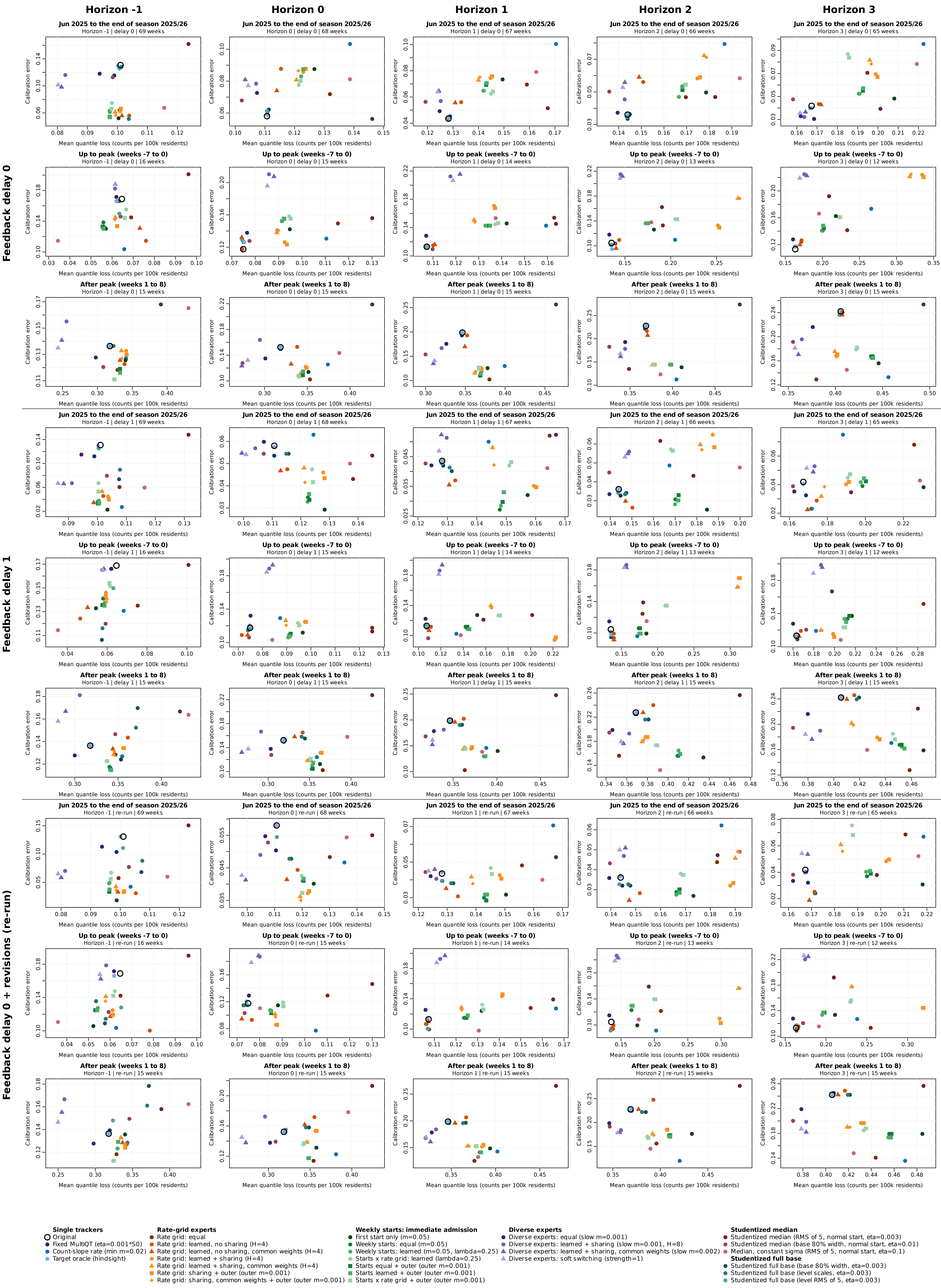
Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.

Horizon -1
Horizon 0
Horizon 1
Horizon 2
Horizon 3
Feedback delay 0
Feedback delay 1
Feedback delay 0 + revisions (re-run)
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks
Horizon 0 | delay 0 | 68 weeks
Horizon 1 | delay 0 | 67 weeks
Horizon 2 | delay 0 | 66 weeks
Horizon 3 | delay 0 | 65 weeks
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks
Horizon 0 | delay 0 | 24 weeks
Horizon 1 | delay 0 | 23 weeks
Horizon 2 | delay 0 | 22 weeks
Horizon 3 | delay 0 | 21 weeks
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 19 weeks
Horizon 0 | delay 0 | 19 weeks
Horizon 1 | delay 0 | 19 weeks
Horizon 2 | delay 0 | 19 weeks
Horizon 3 | delay 0 | 19 weeks
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks
Horizon 0 | delay 1 | 68 weeks
Horizon 1 | delay 1 | 67 weeks
Horizon 2 | delay 1 | 66 weeks
Horizon 3 | delay 1 | 65 weeks
Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 24 weeks
Horizon 0 | delay 1 | 24 weeks
Horizon 1 | delay 1 | 23 weeks
Horizon 2 | delay 1 | 22 weeks
Horizon 3 | delay 1 | 21 weeks
After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 19 weeks
Horizon 0 | delay 1 | 19 weeks
Horizon 1 | delay 1 | 19 weeks
Horizon 2 | delay 1 | 19 weeks
Horizon 3 | delay 1 | 19 weeks
Jun 2025 to the end of season 2025/26
Horizon -1 | re-run | 69 weeks
Horizon 0 | re-run | 68 weeks
Horizon 1 | re-run | 67 weeks
Horizon 2 | re-run | 66 weeks
Horizon 3 | re-run | 65 weeks
Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 24 weeks
Horizon 0 | re-run | 24 weeks
Horizon 1 | re-run | 23 weeks
Horizon 2 | re-run | 22 weeks
Horizon 3 | re-run | 21 weeks
After peak (weeks 1 to 8)
Horizon -1 | re-run | 19 weeks
Horizon 0 | re-run | 19 weeks
Horizon 1 | re-run | 19 weeks
Horizon 2 | re-run | 19 weeks
Horizon 3 | re-run | 19 weeks
Calibration error
Mean quantile loss (counts per 100k residents)
Single trackers
○ Original
● Fixed MultiQT (eta=0.001*S0)
● Count-slope rate (min m=0.02)
● Target oracle (hindsight)
Rate-grid experts
● Rate grid: equal
● Rate grid: learned, no sharing (H=4)
▲ Rate grid: learned, no sharing, common weights (H=4)
● Rate grid: learned + sharing (H=4)
▲ Rate grid: learned + sharing, common weights (H=4)
■ Rate grid: sharing + outer (outer m=0.001)
◆ Rate grid: sharing, common weights + outer (outer m=0.001)
Weekly starts: immediate admission
● First start only (m=0.05)
● Weekly starts: equal (m=0.05)
● Weekly starts: learned (m=0.05, lambda=0.25)
● Starts x rate grid: learned (lambda=0.25)
■ Starts equal + outer (outer m=0.001)
■ Starts learned + outer (outer m=0.001)
■ Starts x rate grid + outer (outer m=0.001)
Diverse experts
● Diverse experts: equal (slow m=0.001)
● Diverse experts: learned + sharing (slow m=0.001, H=8)
▲ Diverse experts: learned + sharing, common weights (slow m=0.002)
▲ Diverse experts: soft switching (strength=1)
Studentized median
● Studentized median (RMS of 5, normal start, eta=0.003)
● Studentized median (base 80% width, normal start, eta=0.01)
● Median, constant sigma (RMS of 5, normal start, eta=0.1)
Studentized full base
● Studentized full base (base 80% width, eta=0.003)
● Studentized full base (level scales, eta=0.003)
● Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | New York (36) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

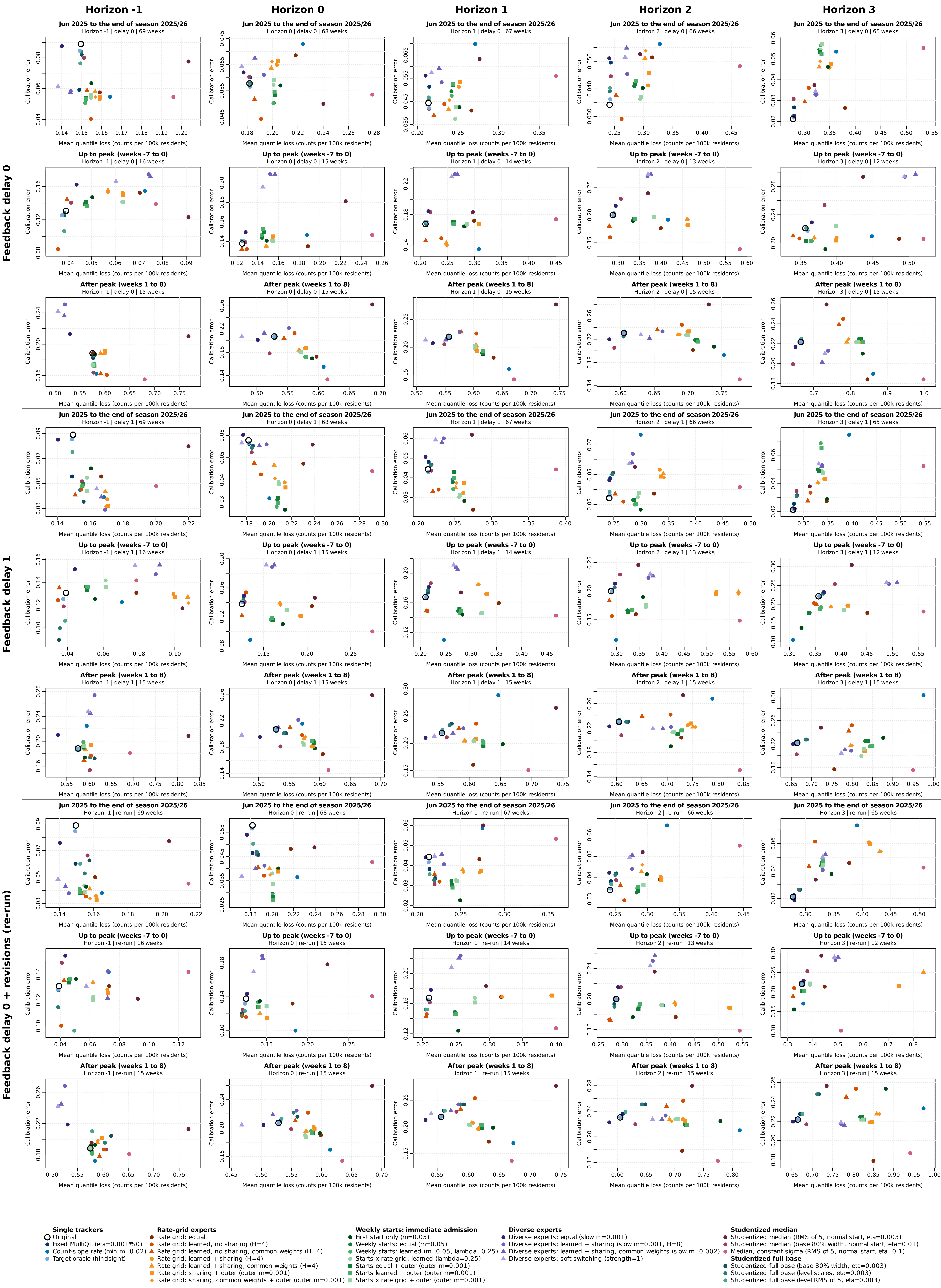
Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Pennsylvania (42) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

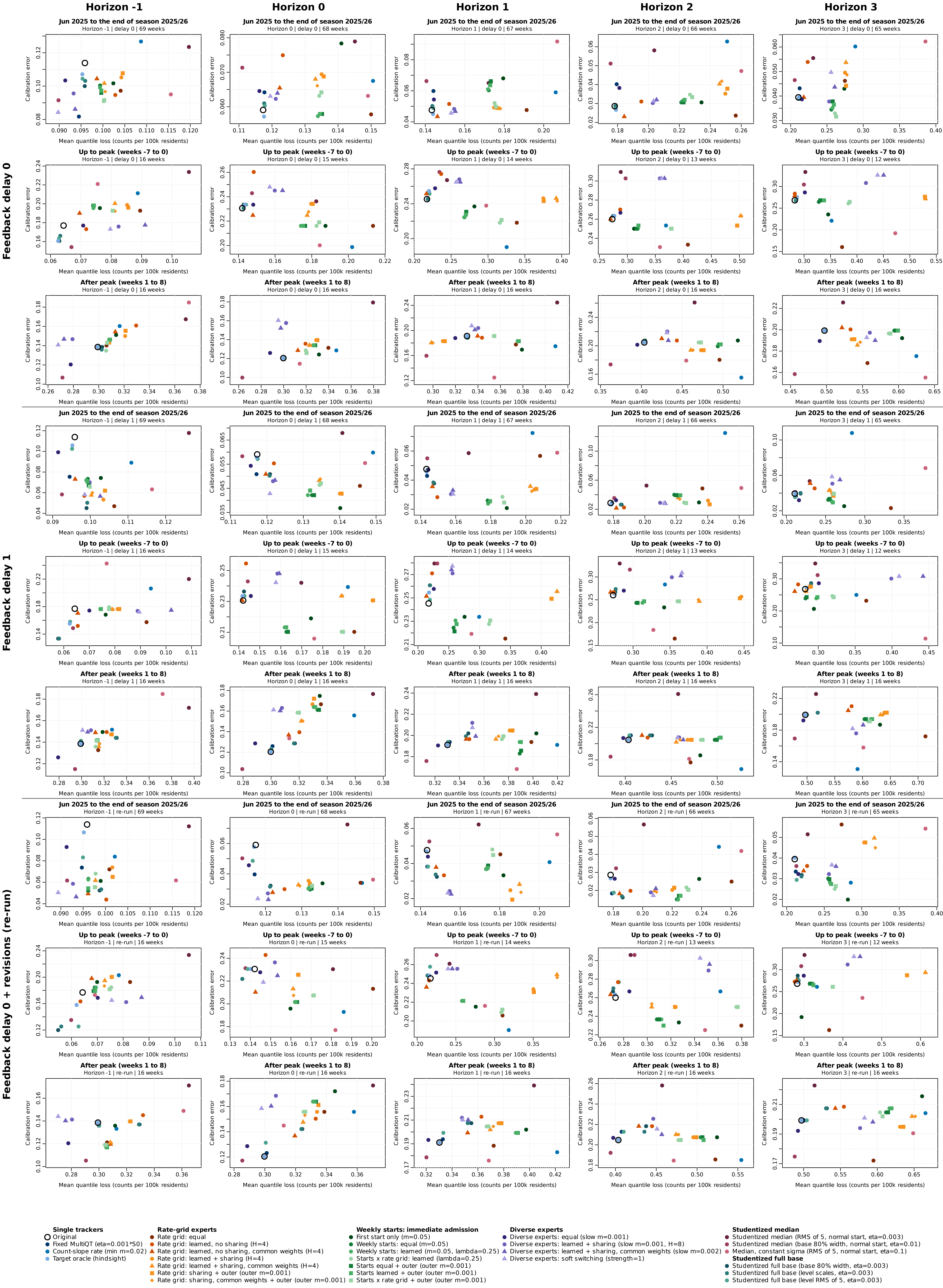
Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Illinois (17) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).